# Public Comment

**Federal Energy Regulatory Commission (FERC) ANOPR on Connecting Large Loads**
**Docket No. RM26-4-000**

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## Qualifications and Introduction

I am Herbert (Herb) Schrayshuen, a licensed professional engineer in the State of New York and a power industry consultant. I offer this technical guidance to the FERC regarding a subset of the matters presented in the ANOPR related to power system resilience and reliability. My resume is appended to these comments for reference. My comments are my own and do not reflect the positions of any of the clients served by my company, Power Advisors LLC.

I respectfully submit these technical comments in response to FERC's ANOPR on the interconnection of large loads to the transmission system related to paragraph 31 principle #14 of the ANOPR. The rapid growth of data centers, cryptocurrency mining, and industrial electrification present substantial resilience and reliability risk to the power system. These risks are best addressed through the augmentation of NERC's existing mandatory Reliability Standards (some of which are Regional Standards) or the addition of new standards applicable to large load analysis and connection facility design to ensure the continued resilience and reliability of the power system.

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## Reliability Risks of Large Load Integration

There have been many comments submitted in the ANOPR proceeding regarding federal/state jurisdiction, ratemaking, retail rate payer cost protection and general equity. While the NERC standards were mentioned in a few cases and many comments call for large load entity adherence to applicable NERC mandatory requirements, there is relatively little comment on the need for augmentation of specific NERC standards. NERC's own comments focus largely on its large load initiative in the technical areas and on setting the stage for registration of large loads, which I also believe is essential to preserve the reliable operation of the power system. Large load entities will likely be transmission owners and will be UFLS entities as currently defined in the NERC PRC-006-5 standard.

NERC mentions the FAC-001 and FAC-002 standards related to establishing interconnection technical requirements and conducting interconnection studies which examine reliability needs of the transmission systems to which the large load will connect. I strongly support modification of all currently applicable and potentially applicable standards to specifically reference interconnection resilience and reliability risks emanating from large load entity connections to the transmission system.

Here is a list of key resilience and reliability concerns which are best addressed via NERC's and the Region's standards processes through augmentation of their respective mandatory requirements and regional criteria.

- **Resilience Concerns:** Inflexible large loads have been demonstrated to reduce the Bulk Electric Systems (BES) ability to withstand disturbances, increasing risks of cascading outages during extreme events
- **Sudden Concentrated Demand:** Large loads can materialize quickly. These loads can materialize in less than the five year annual review cycle for automatic underfrequency load shedding program functional verification under PRC-006-5.
- **Geographic Clustering:** As has already been documented in the Northern Virgina disturbances, concentrated siting of large loads stresses local transmission infrastructure, creating congestion and voltage stability risks.
- **Operational Uncertainty:** Demand profiles of large loads while often characterized by large load advocates as having beneficially high load factors, have also been shown to be volatile over short time intervals and therefore have the ability to significantly complicate balancing authority operations and generating resource reserve adequacy.

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## Needed Improvements in NERC Reliability Standards

What follows is a list of suggested policy actions for the FERC to consider when it issues the Notice of Proposed Rulemaking which will emanate from the ANOPR comments.

1. **Automatic Underfrequency Load Shedding (PRC-006-5)**
    - The presently approved PRC-006-5 requirements focus on ensuring that the Planning Coordinators design automatic UFLS programs to arrest frequency decline after disturbances. In its purpose statement the standard characterizes its requirements as the last resort system preservation measure.
    - The standard (at R4) unfortunately only requires verification of the functionality of the automatic underfrequency load shedding program every five years. This functionality review is typically performed on the basis of the installed load, not projected load. Although in some areas these studies are performed on a specific forward looking "study year" basis. This is a good approach, but not universally required or implemented. It should be. The current requirement is:

        > *"R4 Each Planning Coordinator shall conduct and document a UFLS design assessment at least once every five years that determines through dynamic simulation whether the UFLS program design meets the performance characteristics in Requirement R3 for each island identified in Requirement R2."*

    - Some regions have more prescriptive automatic UFLS supporting requirements. In particular note:
        - PRC-006-NPCC-2 which builds on the NERC PRC-006-5 standard and incorporates compensatory load shedding concepts for situations where certain

generators are non-conforming from a ride-through perspective. See R13 and R15. In NPCC this type of requirement may impact large loads connecting in the NPCC Region that propose to also bring generation to their large load installation.

- NERC should be encouraged to explore this concept as an addition to PRC-006-5.

- PRC-006-SERC - 3 which builds on NERC-PRC-006-5 and provides in its R4 the obligations of UFLS Entities (100 MW or greater)

  *"R4 Each UFLS entity that has a total load of 100 MW or greater in a Planning Coordinator area in the SERC Region shall implement the UFLS scheme developed by their Planning Coordinator. UFLS entities may implement the UFLS scheme developed by the Planning Coordinator by coordinating with other UFLS entities…"*

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- I recommend that the FERC offer policy guidance in its NOPR to promote NERC augmentation of PRC-006-5 as follows:

  - The FERC should direct NERC to make PRC-006 universally forward looking by requiring review and analysis of automatic UFLS program functionality through simulation studies prior to the interconnection and energization of large loads. Taking this step would ensure that automatic UFLS schemes are not compromised by the addition of large concentrated electric demand and that sufficient identified load is available for inclusion in automatic load shedding programs to stabilize frequency in response to system disturbances.
  - Alternatively, the FERC can direct that automatic underfrequency load shedding programs be evaluated on an interval more frequently than the present five-year interval.
  - Review the two regional standards and possibly other less formal regional processes being used to manage automatic underfrequency load shedding programs. Identify opportunities to coordinate and bring the best automatic underfrequency program management practice features into a future version of PRC-006-5.
  - Consider as a matter of policy that large loads do to their size should become UFLS Entities as the term in used within the PRC-006-5 standard because of their tremendous impact on the load situation for the connecting Transmission Owner.

    *UFLS entities shall mean all entities that are responsible for the ownership, operation, or control of UFLS equipment as required by the UFLS program established by the Planning Coordinators. Such entities may include one or more of the following: 4.2.1 Transmission Owners 4.2.2 Distribution Providers 4.2.3 UFLS-Only Distribution Providers*

- No matter which policy path is pursued, automatic underfrequency load shedding programs must be analyzed on a forward looking basis, taking into account the presence of large loads, rather than on an “as connected basis” after electric service facilities are designed, committed and energized to service the large load.

  - While recognizing that not all large load projects are certain to be built, and to facilitate the forward looking analysis, uniform inclusion rules must be developed by planning coordinators so that large load projects and their impact on automatic underfrequency load shedding programs can be analyzed with confidence. It is best also, to the extent possible, to conduct the analysis in clusters when large load projects are concentrated in portions of the transmission system.

FERC proposals or directives for action of this kind in its NOPR will ensure that automatic UFLS programs are fully analyzed and most importantly are not compromised by the unstudied addition of concentrated demand. The studies will set the stage to assure that sufficient load is placed under automatic UFLS program control to achieve the resilience goal of stabilizing frequency and arresting frequency decline after a system disturbance.

2. **Transmission Planning (TPL-001-5.1)**

A fundamental principle of prudent power system engineering is that the power system should not be placed into an unstudied state. There are a number of mandatory NERC standards that embody this principle, but TPL-001-5.1 is at the core of implementing this approach from a system planning perspective.

I recommend that the FERC take action in its NOPR to:

- Explore the need for augmentation of the TPL-001-5.1 to require explicit modeling of large, concentrated loads in both near term and long term planning assessments even if those projects are offering to take non-firm “flexible” electric service.
- While recognizing that not all large load projects are certain to be built, uniform inclusion rules must be developed so that the large load project’s impact on power deliverability can be analyzed with confidence regarding their likely materialization and in clusters if concentrated on portions of the transmission system, rather than individually.
- Mandate sensitivity studies that stress-test transmission system planning scenarios involving simultaneous large load growth and sudden large load loss under the appropriate transmission and generation contingencies identified in Table 1 of the standard.
- Direct NERC to modify Table 1 contingencies if they are not adequate to the purpose of testing the transmission system for large load operating behavior.
- Direct NERC to consider whether amendments to footnote 12 and to Attachment 1 of Table 1 of TPL-001-5.1 are needed to accommodate large load projects willing to be “flexible”, i.e., willing to take non-firm electricity service. This will allow utility planning processes to adapt and provide service flexible large loads willing to take non-firm service in a way that complies with the standard. As it stands, offering non-firm transmission service may not meet the requirements in TPL-001. Under the current requirements, it is not permitted to plan the transmission system to provide electric service under TPL-001-5.1 on a non-firm basis (i.e., permitting non-consequential load loss).

3. **Interconnection Process and Operational Requirements**

   The interconnection process is where resilience and reliability risks should be identified as early as possible for the benefit of the existing electricity customers and for the benefit of the new large load customers. To make sure that the risks are surfaced early enough to take prudent engineering and design action, the following actions by the FERC are recommended for consideration:

   - The FERC should consider taking action to:
     - Require all regional transmission organizations, planning coordinators and transmission owners to establish minimum technical interconnection requirements for connecting large loads, similar but not identical to generator requirements (unless the new large load bringing its own generation to the process), including voltage support, frequency ride-through, and demand response capability. These directives would be best applied in future versions of NERC Standards FAC-001 -4 and FAC-002-4 as also noted in NERC's ANOPR comments at page 4.
     - Authorize NERC to register large load entities as Transmission Owners and Transmission Operators if the new large load interconnecting facilities qualify as BES Elements and are not turned over to the incumbent connecting Transmission Owners and Transmission Operators for operation and maintenance after commissioning.
     - For situations where new large load entities are proposing to implement Remedial Action Schemes (RAS) to possibly avoid FERC jurisdiction, to require that such large load sponsored RAS based solutions be designed (or at least reviewed subject to approval) owned and operated by existing connecting Transmission Owner to bring such RAS within the scope of PRC-012-2 for RAS design and PRC-017-1 for RAS maintenance.

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## Alignment with NERC Large Load Task Force Findings

Since the FERC is aware of NERC's Large Load Initiative as described in NERC's ANOPR filing, take note of NERC's characterization of the significant risk to resilience and reliability the may result from large loads.

For reference and at a high level the **NERC Large Load Task Force (LLTF)**, in its April 2025 report, highlighted several critical points:

- *"UFLS program adequacy must be reassessed in light of large load interconnections, as existing schemes may not provide sufficient frequency stabilization."*
- *"Large loads are materializing faster than traditional planning processes can adapt."*
- *"Existing Reliability Standards do not explicitly address the unique risks posed by concentrated demand growth."*
- *"Enhanced coordination between load developers, transmission planners, and balancing authorities is essential to mitigate reliability risks."*

These findings by NERC reinforce the need for the FERC to direct NERC to revise PRC 006, TPL-001, and other standards as appropriate to explicitly incorporate power system resilience and reliability preservation considerations in utility planning. There is an urgent need for large load integration and operations process modification. At a minimum changes and augmentation of the processes governed by PRC-006 and TPL-001 are necessary. Other standards which support these standards will also require attention as necessary[1].

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**Recommendations for FERC Oversight of Large Load Interconnection Processes**

Ongoing FERC oversight of large load resilience and reliability risks to the transmission system will be needed to monitor the significant risk to resilience and reliability posed by incorrectly planned large load additions.

To ensure timely action to stay "ahead of the curve" with respect to the forgoing resilience and reliability risks, I recommend that the FERC consider:

- Directing that NERC report annually (or at an interval suitable to the FERC and NERC) to the FERC on the adequacy[2] of manual and automatic UFLS programs and resilience planning in each US Interconnection.
- Direct NERC to initiate a process to develop standards authorization requests and file modified or new standards responsive to the resilience and reliability concerns identified above within a time frame of FERC's choosing (suggested to be two years or less) to assure reliable and resilient large load integration that does not compromise the reliable operation of the BES.

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## Conclusion

The integration of large loads represents a transformative challenge to the resilience and reliability of the power system. Without augmented and strengthened NERC Reliability Standards, particularly PRC-006, and TPL-001, the resilience and reliability of the power delivery system will be compromised with the addition of unprecedented numbers of large load projects.

FERC should act decisively in its NOPR to ensure that mandatory reliability standards evolve <u>ahead</u> of these emerging risks, safeguarding power system resilience and reliability. FERC through its nationwide responsibility under Section 215 of the Federal Power Act, must at a minimum deliver continued resilience and reliability of the power system for all users to meet the expectations outlined on NERC's 2013 informational filing with the FERC regarding the planning and operational outcomes that are expected in order to deliver an "adequate level of reliability"[3].

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1 Additional standard families which may be impacted include MOD, FAC, EOP, TPL  and other PRC standards besides PRC-006.
2 See NERC Informational filing regarding on the Definition of "Adequate Level of Reliability" NERC Informational Filing in RR06-1-000 May 10, 2013.
3 See filing referenced in footnote 2, in particular performance objective #4.

The power system will quickly reveal if the FERC, NERC, registered entities get it wrong through outages that are more extensive, exhibit more lengthy recovery times, and are more damaging to hard to replace power delivery equipment. Such outages run the risk of additional loss of life, way beyond what was experienced in Texas during Winter Storm Uri.[4] Such unacceptable outcomes can be mitigated with proper engineering foresight, planning foresight and by requiring thoughtful forward looking technical review and analysis of the connection of the proposed large loads prior to design, construction and energization.

The FERC should consider the forgoing comments in the interest of public health and safety when crafting the Notice of Proposed Rulemaking addressing large load interconnection. While the above comments may seem like an expansion of NERC rules into the load sector which has been primarily the province of the US States, the fact remains that large city size loads connected to the transmission system have a direct impact on the resilience and reliability of the interconnected North American power systems.

Thank you in advance for your attention and thoughtful consideration to the forgoing comments in the development of the Large Load NOPR.

**Respectfully submitted,**
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[4] Winter Storm Uri official death toll report-Texas

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**Current Title & Activity:** Since February 2013, Principal, Power Advisors, LLC providing consulting services to the electric utility industry with experience in:

- Utility asset management
- Utility power system planning and reliability
- Utility reliability regulation including NERC standards compliance related matters
- Regulatory matters and dispute resolution

**Prior Titles and Activities:**

VP & Director Reliability Assessment and Performance Analysis (2012-2013) - North American Electric Reliability Corporation (NERC)
VP & Director Standards & Training (2010-2012) – NERC
Director Reliability Assessment – SERC (2008-2010),
VP Reliability Compliance – National Grid (2006-2007),
VP Transmission Commercial Services – National Grid (2002-2006)
Director System Electric Assets – Niagara Mohawk Power Corporation (1999-2002)
Previous – numerous roles of increasing accountability within the managerial, technical, and engineering functions of Niagara Mohawk Power Corporation and American Electric Power.

**Relevant Experience:** Over 40 years' experience in the electric power industry including engineering and regulatory roles for the industry including reliability regulation under NERC mandates. Managed technical groups in disciplines including, engineering, reliability regulations, asset management, transmission and distribution engineering, transmission planning, project management, transmission account management, and transmission siting. Provided expert testimony in regulatory and civil legal forums, settlements and transaction design. Extensive experience in generation, transmission and distribution system operations, power transactions, planning and asset health issues. Leadership roles in many FERC and state regulatory proceedings, disputes resolution and policy making activities. Two examples are coordination of National Grid's interrogatory responses to the Eastern Interconnection Blackout of 2003 and participation in the development and implementation of the Energy Policy Act of 2005 with respect to electric industry reliability regulation leading to the formation of the Electric Reliability Organization (ERO) as NERC.

**Education & Training:**

Masters Business Administration – Rensselaer Polytechnic Institute
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